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Phishing Email's Detection Using Machine Learning and Deep Learning

Nishant Santosh Paradkar
M.Sc. Computer Science
SVKM's Mithibai College of Arts Chauhan Institute Of Science And Amrutben Jivanlal College Of Commerce And Economics
Mumbai, India
nishant.paradkar2000@gmail.com

***Abstract*—In today's world, all of us are dependent on emails. Emails are a very efficient and fast way of sending a message to someone. But malicious users often use it to send fraudulent emails with fake links that steal user credentials like credit card details, login-id, passwords, etc. These emails are called phishing emails. These emails constitute identity fraud as the emails are interpreted to be from banks or other multinational companies. Many existing solutions require the user to check for grammar errors, check the email-id, or avoid clicking any links. But all these actions require human involvement. In this paper, I have implemented and compared current Machine Learning and Deep Learning techniques used with Natural Language Processing to detect phishing emails and achieved an accuracy of 98%.**

***Keywords*—Phishing, Machine Learning, Deep Learning, Bi-LSTM, Convolutional Neural Network.**

I. Introduction

Phishing comes in the category of social engineering attack which constitutes identity fraud. Phishing emails contain hidden links in text messages and these messages usually resemble a sense of urgency i.e. "Your account will be blocked if you don't respond within 30 mins". These hidden links are malicious links that redirect the user to a fake site. On clicking the link, the user's IP is revealed to the hacker, and a javascript running on that malicious website may be executed on the user's system like stealing cookies, saved passwords, etc. Also, if the user shares his login details on the form, those are revealed as well. One of the biggest hazards in the world today is phishing email, which results in enormous financial losses. Additionally, the number of phishing emails has grown alarmingly in recent years. According to the Anti-Phishing Working Group [1] 1,270,883 phishing attacks were observed in the 3rd quarter of 2022.

To combat the threat of phishing emails, more effective phishing detection technology is required. As mentioned in [2] current traditional methods are inefficient and require human involvement. Hence a machine learning technique would be highly useful to remove human involvement. The problem is text classification as we need to classify phishing emails and benign emails. In this paper, I have tried different machine learning and deep learning techniques to solve the text classification problem and presented a comparative analysis of the same.

II. Related Work

In [2] the authors used SVM, Biased SVM, Leave One model out and k-means clustering to train on the SpamAssassin dataset. Their experiment concluded that SVM and Artificial Neural Networks gave better results than K-Means clustering, Leave One Model Out, and Self-Organizing Maps. In [3] SVM, KNN, Random Forest Classifier, and Logistic Regression were used for text classification. Their experiment concluded that Random Forest Classifier performed better than Logistic Regression, Naive Bayes, SVM, and Decision Tree. In [4] the authors used Naive Bayes model and Random Forest Classifier and achieved an accuracy of around 98% and 92% respectively. In [5] the authors used the Long Short-Term Memory (LSTM) model and achieved an accuracy of 96.64%. The author also used a CNN Model and achieved an accuracy of 95.97%. Their experiment showed that a Deep Learning model performs better than Machine Leaning models for text classification. In [6] the authors combined Naive Bayes and Decision Tree and achieved an accuracy of around 96.77% for both models.

III. Methodology

A. Dataset

The dataset is taken from [7] ENRON Corpus which contains over 500,000 emails. This dataset is a sample set of 20000 phishing and benign emails retrieved from the ENRON Corpus. This dataset contains 8336 Phishing emails and 11664 Benign emails.

B. Data Preprocessing

Here the dataset was stripped of all stop words, null entries, punctuations, etc. Then the text was tokenized using the Keras tokenizer. After tokenization the data is uneven and a Machine Learning and Deep Learning model requires uniform data, so the dataset was then padded using Keras pad sequences. The padding method adds 0 at the start or end of the dataset to maintain uniformity. After that I used the keras LabelEncoder to convert the labels to 0's and 1's. The 0's represented benign emails and 1's represented Phishing emails. Then the data was divided into training and testing at 75% and 25% respectively.

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C. Techniques Used

- Logistic Regression: This is a machine-learning model which is used for binary classification. This model estimates the probability of an event occurring or not.
- Naive Bayes: The Bayes theorem is the foundation of the Naive Bayes classification algorithm. Naive Bayes is widely used for text classification.
- Decision Tree Classifier: The Decision Tree Classifier is a machine-learning technique widely used to solve problems like classification and regression.
- Support Vector Machine: SVM is a machine learning algorithm that is used to solve problems of classification and regression. SVM comes under the banner of supervised learning.
- Long Short-Term Memory: Long Short-Term Memory (LSTM) comes in the category of a recurrent neural network. As LSTM has feedback connections, it can process large data sequences simultaneously.
- Bidirectional-LSTM: Bidirectional LSTM consists of two LSTM models one takes input in the forward direction and the other takes input in the backward direction. This basically reverses the input data and learns again which provides better results.
- Convolutional Neural Networks: CNN are deep learning networks used for image classification. If applied with an embedding layer, CNN can be used for text classification.

D. Model

In the LSTM model, an embedding layer was built with 1000-word embeddings. The embedding layer converts each word into a fixed-length vector. This embedding layer is passed to the LSTM layer. In the LSTM layer, I used a dropout of 0.2 and a recurrent dropout of 0.2 to avoid overfitting. Dropout drops a certain number of neurons that are not useful. After that, a single Dense layer was used with a sigmoid activation function for binary classification. The loss function used was binary cross-entropy as it is best for binary classification and adam optimizer was used. This model had 1,021,501 trainable parameters. I specified 5 epochs for the LSTM model. It took 2.5 hours for the model to train.

In the BI-LSTM model, the same specifications were used as that of the LSTM model but the LSTM layer was wrapped with a Bi-Directional layer. The Bi-Directional layer creates two LSTM layers, one with forward input and the second with backward input. This creates more data by using the same data but in reverse. This is a type of data augmentation method. Data augmentation is the process of creating dummy data based on available data to better train the model as it is exposed to a variety of data samples. This helps to increase the accuracy of the model. This model had 1,743,001 trainable parameters. The training took 5 hours to complete on 5 epochs. In the CNN model, I use an embedding layer to create word embeddings and to create 3-dimensional input data as CNN requires 3-dimensional input. Here I used a conv1d layer. After that, I added a 1-dimensional max-pooling layer which helps in reducing parameters to learn by returning the maximum values of a given input matrix. Then I used a flatten layer to convert the output from 3 dimensions to 2 dimensions. After that, a single layer with 250 neurons and relu activation function is used and lastly, a sigmoid activation layer is applied. The loss function used was binary cross-entropy as it is best for binary classification and adam optimizer was used. This model had 851,605 trainable parameters. The training took 10 minutes to complete with 50 epochs.

IV. Results & Conclusion

In machine learning, first I tried training the data on the Logistic Regression model. I received an accuracy of 57.91% on testing data. Next, I tried the Naive Bayes model where I received an accuracy of 54.66% on testing data. Using Decision Tree Classifier, I got an accuracy of 73.27% on testing data. After training the data on Support Vector Machine I was able to receive an accuracy of 69.02%. All the used machine learning techniques are underfitting on the provided dataset. This proves that classical machine-learning techniques are not efficient for text classification.

In deep learning, I tried the Long Short-Term Memory (LSTM) model. The training took 2.5 hours and it achieved an accuracy of 97.32% on testing data. This model had a F1-Score of 0.9786 which is a good score as we don't want false positives or false negatives because not detecting a phishing email and detecting a benign email as phishing are equally catastrophic. After that, I used a Bi-directional LSTM model. This model took 5 hours to complete training and an accuracy of 98.04% was achieved. This model had a F1-Score of 0.9825. The validation loss in LSTM was 0.0923 whereas in Bi-LSTM it was 0.0609 which is relatively less than LSTM. After that, I used a CNN model. The training took 10 minutes which is very less compared to LSTM and Bi-LSTM. Even with less training time the CNN model was able to achieve an accuracy of 98.05% and a validation loss of 0.2138. This model could achieve a F1-Score of 0.9826. Loss in the CNN model was more than that of LSTM and Bi-LSTM. The CNN model is a tradeoff between speed and accuracy. Comparing the F1-Scores of all three model's, I conclude that CNN model is a more viable option as it achieved nearly the same accuracy as that of the Bi-LSTM model with less training time and a good F1-Score. Only drawback is that loss is a little higher.

TABLE I. ACCURACY OF DIFFERENT MODELS.

Methods Used	Training Accuracy	Testing Accuracy
Logistic Regression	59.31%	57.91%
Support Vector Machine	83.27%	69.02%
Decision Tree Classifier	99.95%	73.23%
Naive Bayes	55.57%	54.66%
LSTM	98.65%	97.32%
Bi-LSTM	98.93%	98.04%
CNN	99.93%	98.05%

From the obtained results shown in Table I we observed that Machine Learning techniques have less trainable parameters as compared to deep learning techniques. machine learning

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techniques take less time to train than deep learning techniques but even though deep learning techniques use more resources still they provide better results than machine learning techniques. These results prove that deep learning techniques are more suitable for text classification than machine learning techniques.

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